SmartDAQ: Intelligent Data Acquisition System Enables Low-latency Photoacoustic Imaging System

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***Abstract*—Photoacoustic imaging (PAI) stands at the cutting edge of biomedical imaging but faces significant challenges due to fixed data acquisition parameters and high latency in data transmission. These constraints curtail system adaptability and prolong data processing times, thereby hindering real-time applications such as intraoperative navigation due to the protracted transfer of raw data to CPUs or GPUs for image reconstruction. In this paper, we present SmartDAQ, an innovative intelligent data acquisition system that incorporates FPGA chips to enable dynamic reconfiguration of acquisition parameters and in-system PA image reconstruction. This groundbreaking approach significantly enhances system programmability and drastically reduces data transmission latency, thus making low-latency, real-time PAI systems feasible. We validated the smartDAQ system through in vivo experiments on a human finger, confirming its real-time imaging capabilities at 20 fps and its support for multiple algorithms, delivering high-quality images. These results underscore its potential for real-time clinical applications.**

Keywords—Photoacoustic imaging, data acquisition, SmartDAQ, FPGA, low-latency, real-time

I. Introduction

Photoacoustic imaging (PAI) [1], [2] is an emerging modality that synergistically combines the optical contrast with the deeper penetration depth of ultrasound. Exploiting the photoacoustic effect, where biological tissues, upon exposure to nanosecond laser pulses, absorb optical energy, undergo thermoelastic expansion, and emit ultrasonic waves [3], PAI offers unrivaled imaging capabilities.

PA systems that are configurable, portable, low-latency, and capable of real-time, high-quality imaging are particularly desirable for outdoor applications and clinical scenarios such as intraoperative navigation. These advanced systems provide prompt and precise delineation of critical structural and functional information of target tissues, thereby significantly enhancing diagnostic and therapeutic decision-making. While significant advancements have been made in optimizing backend image reconstruction algorithms to expedite imaging processes, challenges still remain. Traditional beamforming techniques have been refined to facilitate real-time, high-quality imaging [4], [5], and deep learning methods have been employed to empower real-time PAI systems [6]-[8]. The former leverages CPU processing, whereas the latter utilizes GPUs for neural network training. Both approaches, however, necessitate the transfer of data from acquisition cards to backend computing units, thereby introducing substantial data transmission latency and limits the overall efficacy of data processing. Furthermore, the deployment of CPUs and GPUs significantly escalates power consumption and increases system size. Efforts have also been made to harness Field-Programmable Gate Arrays (FPGA) for accelerating image reconstruction. Some approaches utilize FPGAs for controlling data acquisition [9], [10], thereby achieving programmable acquisition workflows without mitigating high transmission latency. Other methods involve the direct implementation of reconstruction algorithms on FPGAs [11]-[13], wherein PA data from acquisition cards is channeled into the FPGA for image reconstruction. Despite accelerating the reconstruction process, these methods fail to resolve issues related to fixed acquisition configurations and high latency, as the FPGA still functions primarily as a backend computing unit.

In this paper, we introduce SmartDAQ, an advanced and intelligent data acquisition system that leverages the formidable data processing, reconfigurability, and high-speed communication capabilities of an embedded FPGA. By integrating the FPGA chip directly into the acquisition card, SmartDAQ not only enables comprehensive control over the data acquisition process and highly flexible parameter configuration , but it also performs real-time PA image reconstruction within the acquisition system. This approach significantly reduces data transmission latency and enhances

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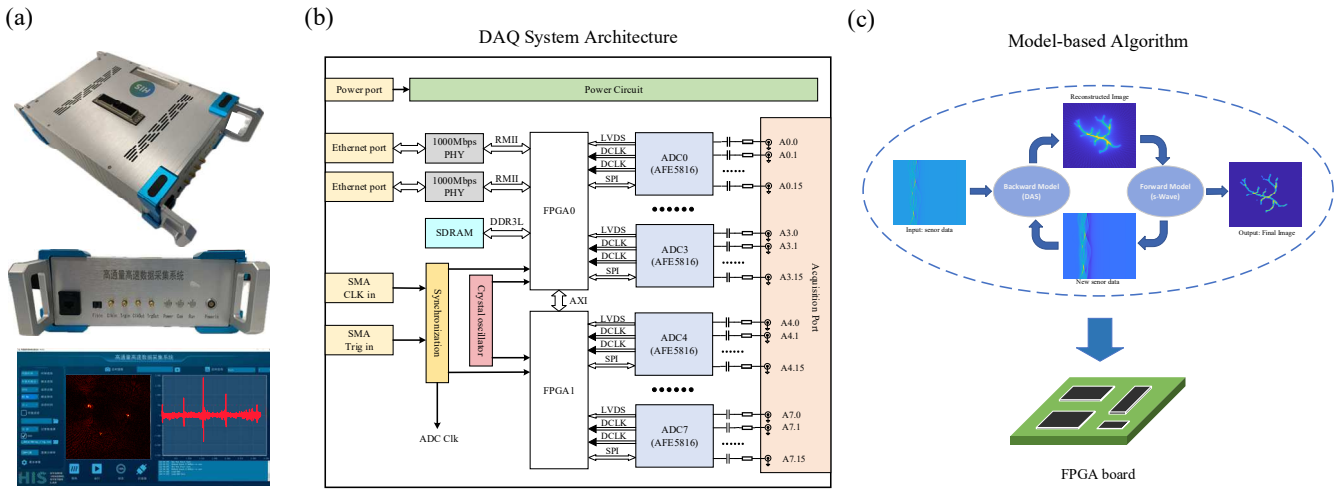


Fig. 1. Design of smartDAQ. (a) The exterior and GUI interface design of smartDAQ. (b) Hardware architecture of the smartDAQ system. (c) Schematic of Model-based algorithm implementation within FPGA.

the flexibility of data acquisition. The feasibility and clinical potential of SmartDAQ were validated through in vivo imaging experiments on a human finger, demonstrating its capability for real-time, high-quality imaging.

II. Method

A. The Hardware Design of SmartDAQ

As depicted in Fig. 1(a), the data acquisition system can be broadly categorized into software interface design and hardware circuit design. We employed QT to develop a versatile software interface that provides comprehensive configurability for acquisition parameters. Users can select between internal and external clock sources, choose among internal electric trigger, external electric trigger, or external optical trigger modes, and set sampling point ranges, trigger frequency, and sampling duration. The interface also facilitates the activation or deactivation of filtering algorithms and the selection of image reconstruction algorithms. Additionally, it allows for the adjustment of analog-to-digital converter (ADC) settings and amplification factors. The system supports dual graphical display options: one for visualizing single-channel signals and another for displaying reconstructed photoacoustic images.

As depicted in Fig. 1(b), the hardware architecture of SmartDAQ encompasses key functional modules: the power supply, clock generation, ADC acquisition, FPGA control, and communication interfacing. The system operates on a 220V AC supply with a power budget capped at 150W. Internal clocking is supported by an oscillator, and synchronization is further enhanced through external clock inputs and an optical trigger essential for photoacoustic imaging applications. The ADC module leverages the AFE5816 chipset, providing selectable 12-bit/16-bit resolution. Signals acquired via external terminals are digitized and fed into the FPGA, which orchestrates the entire data acquisition process. The FPGA not only executes signal processing and image reconstruction algorithms but also manages on-chip communication using the Advanced Extensible Interface 4 protocol (AXI4) and external data transfer via an Ethernet interface. A critical aspect of the PAI system is the efficient handling of large data volumes. To address this, we incorporate Double Data Rate 3 (DDR3) Static Random-Access Memory (SRAM), enabling high-speed data storage and retrieval, thus ensuring optimal performance during acquisition and image processing tasks. Fig. 1(c) illustrates the FPGA's comprehensive role in consolidating data processing workflows and ensuring robust communication channels within SmartDAQ.

B. Hardware Implementation of Photoacoustic Image Reconstruction Algorithms

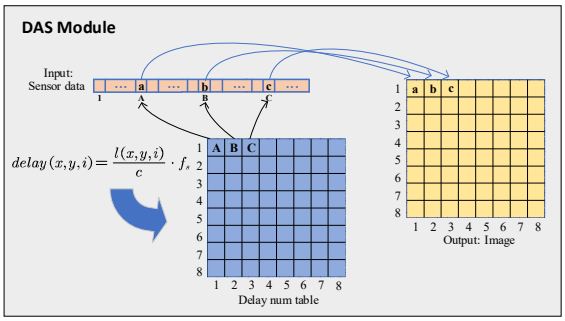


Fig. 2. Hardware Implementation of DAS beamforming algorithm.

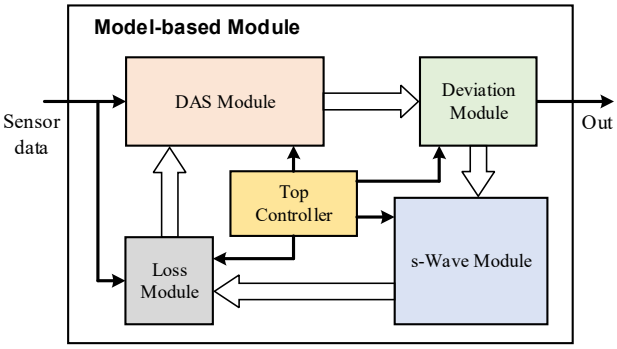


Fig. 3. Hardware Implementation of Model-based algorithm.

In addition to the traditional DAQ design, we have leveraged the programmability of FPGA to implement the PA image reconstruction algorithm. As illustrated in Fig. 2, we have developed the hardware architecture for the classical delay-and-sum (DAS) beamforming algorithm. The time delays required for DAS can be pre-computed and stored in the FPGA's on-chip ROM in advance. Once the acquired PA data is input into the FPGA, the delay data can be fetched

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from the Read Only Memory (ROM) in the corresponding sequence to address the sensor data iteratively into a new RAM. After processing the sensor data from all channels, the final reconstructed image can be output from this RAM.

Furthermore, we present an advanced model-based algorithm [14], meticulously implemented in hardware as depicted in Fig. 3, wherein the transition from signals to images is accomplished utilizing DAS as the backward model, while the superposed Wave (s-Wave), as the forward module, efficiently handles the inverse process of converting images into signals. This iterative mechanism is initiated by juxtaposing the new sensor data, synthesized by the s-Wave module, against the initially acquired raw sensor data; the resultant discrepancy is quantified and employed to generate a residual image through DAS. Subsequently, this residual image is weighted and superimposed on the image from the previous iteration. This iterative refinement persists, systematically enhancing the reconstruction, until the discrepancy between the sensor data converges within an acceptable threshold, culminating in the output of the final reconstructed image. Detailed implementation can be found in our previous work [15].

III. Experiments and Results

In our in vivo experiment, we employed a 128-element circular transducer with a radius of 30 mm, interfaced with the DAQ system for the acquisition of PA signals from a human finger. The external optical trigger was utilized with a sampling rate of 40 MSPS. The final reconstructed image, with a resolution of 128 × 128 pixels and a physical size of 20 mm × 20 mm, is illustrated in Fig. 4, showcasing the detailed PA imaging of the human finger.

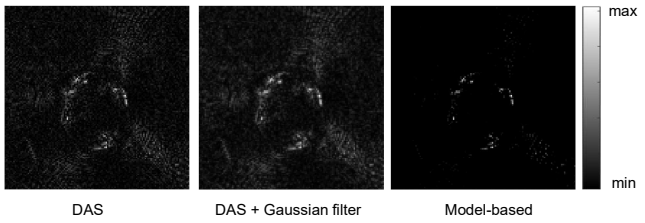


Fig. 4. Reconstruction results of SmartDAQ. The images, shown from left to right, correspond to DAS, DAS + Gaussian filter, and Model-based reconstruction algorithm.

We have implemented the DAS, DAS with Gaussian filtering, and Model-based algorithm within the FPGA. The corresponding results are illustrated in Fig. 4. The cross-sectional images of a human finger clearly highlight blood vessels, which appear as bright spots in the images. The DAS image exhibits noticeable speckle noise and significant artifacts surrounding the finger. Applying Gaussian filtering to the DAS image enhances smoothness and reduces noise. In contrast, the Model-based algorithm effectively suppresses most of the surrounding noise while maintaining the clarity and high pixel intensity of the blood vessels. SmartDAQ achieves a maximum imaging speed of 20 frames per second (fps), making it highly suitable for various real-time imaging applications.

IV. Discussion and Conclusion

In this work, we introduce an advanced intelligent data acquisition system, SmartDAQ, which not only supports the configurability of multiple parameters during data acquisition but also maps diverse image reconstruction algorithms (DAS, Model-based) onto hardware circuits within the system's FPGA. This integration significantly enhances the flexibility, imaging performance, and reconstructed image quality of photoacoustic systems. The versatility and efficacy of SmartDAQ were validated through in vivo imaging experiments on a human finger, demonstrating its capability for flexible data acquisition and rapid, high-quality image reconstruction.

Acknowledgment

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